```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
print(df.head(10))
```

```
2  9305-CDSKC      1  United States  California  Los Angeles     90006
3  7892-POOKP      1  United States  California  Los Angeles     90010
4  0280-XJGEX      1  United States  California  Los Angeles     90015
5  4190-MFLUW      1  United States  California  Los Angeles     90020
6  8779-QRDMV      1  United States  California  Los Angeles     90022
7  1066-JKSGK      1  United States  California  Los Angeles     90024
8  6467-CHFZW      1  United States  California  Los Angeles     90028
9  8665-UTDHZ      1  United States  California  Los Angeles     90029

                Lat Long   Latitude   Longitude  Gender  ...        Contract  \
0  33.964131, -118.272783  33.964131 -118.272783    Male  ...  Month-to-month
1   34.059281, -118.30742  34.059281 -118.307420  Female  ...  Month-to-month
2  34.048013, -118.293953  34.048013 -118.293953  Female  ...  Month-to-month
3  34.062125, -118.315709  34.062125 -118.315709  Female  ...  Month-to-month
4  34.039224, -118.266293  34.039224 -118.266293    Male  ...  Month-to-month
5  34.066367, -118.309868  34.066367 -118.309868  Female  ...  Month-to-month
6   34.02381, -118.156582  34.023810 -118.156582    Male  ...  Month-to-month
7  34.066303, -118.435479  34.066303 -118.435479    Male  ...  Month-to-month
8  34.099869, -118.326843  34.099869 -118.326843    Male  ...  Month-to-month
9  34.089953, -118.294824  34.089953 -118.294824    Male  ...  Month-to-month

  Paperless Billing              Payment Method  Monthly Charges Total Charges  \
0               Yes                Mailed check            53.85        108.15
1               Yes            Electronic check            70.70        151.65
2               Yes            Electronic check            99.65         820.5
3               Yes            Electronic check           104.80       3046.05
4               Yes   Bank transfer (automatic)           103.70        5036.3
5                No     Credit card (automatic)            55.20        528.35
6               Yes            Electronic check            39.65         39.65
7                No                Mailed check            20.15         20.15
8               Yes            Electronic check            99.35       4749.15
9                No            Electronic check            30.20          30.2

  Churn Label Churn Value Churn Score  CLTV  \
0         Yes           1          86  3239
1         Yes           1          67  2701
2         Yes           1          86  5372
3         Yes           1          84  5003
4         Yes           1          89  5340
5         Yes           1          78  5925
6         Yes           1         100  5433
7         Yes           1          92  4832
8         Yes           1          77  5789
9         Yes           1          97  2915

                             Churn Reason
0              Competitor made better offer
1                                     Moved
2                                     Moved
3                                     Moved
4             Competitor had better devices
5  Competitor offered higher download speeds
6               Competitor offered more data
7               Competitor made better offer
8             Competitor had better devices
9             Competitor had better devices

[10 rows x 33 columns]
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
print(df.tail(10))
```

```
7035  9767-FFLEM      1  United States  California       Westmorland
7036  8456-QDAVC      1  United States  California       Winterhaven
7037  7750-EYXWZ      1  United States  California      Yucca Valley
7038  2569-WGERO      1  United States  California           Landers
7039  6840-RESVB      1  United States  California          Adelanto
7040  2234-XADUH      1  United States  California             Amboy
7041  4801-JZAZL      1  United States  California      Angelus Oaks
7042  3186-AJIEK      1  United States  California      Apple Valley



7034     92278  34.457829, -116.139589  34.457829 -116.139589    Male  ...
```

```
7035     92281    33.03679, -115.60503  33.036790 -115.605030     Male  ...
7036     92283  32.852947, -114.850784  32.852947 -114.850784     Male  ...
7037     92284  34.159534, -116.425984  34.159534 -116.425984   Female  ...
7038     92285  34.341737, -116.539416  34.341737 -116.539416   Female  ...
7039     92301  34.667815, -117.536183  34.667815 -117.536183     Male  ...
7040     92304  34.559882, -115.637164  34.559882 -115.637164   Female  ...
7041     92305     34.1678, -116.86433  34.167800 -116.864330   Female  ...
7042     92308  34.424926, -117.184503  34.424926 -117.184503     Male  ...

           Contract Paperless Billing             Payment Method  \
7033  Month-to-month               Yes               Mailed check
7034        One year                No    Credit card (automatic)
7035  Month-to-month               Yes    Credit card (automatic)
7036  Month-to-month               Yes  Bank transfer (automatic)
7037        One year                No           Electronic check
7038        Two year               Yes  Bank transfer (automatic)
7039        One year               Yes               Mailed check
7040        One year               Yes    Credit card (automatic)
7041  Month-to-month               Yes           Electronic check
7042        Two year               Yes  Bank transfer (automatic)

      Monthly Charges Total Charges Churn Label Churn Value Churn Score  CLTV  \
7033            20.05          39.25          No           0          80  5191
7034            60.00         3316.1          No           0          71  4212
7035            69.50        2625.25          No           0          35  4591
7036            78.70         1495.1          No           0          20  2464
7037            60.65          743.3          No           0          24  3740
7038            21.15         1419.4          No           0          45  5306
7039            84.80         1990.5          No           0          59  2140
7040           103.20         7362.9          No           0          71  5560
7041            29.60         346.45          No           0          59  2793
7042           105.65         6844.5          No           0          38  5097

     Churn Reason
7033          NaN
7034          NaN
7035          NaN
7036          NaN
7037          NaN
7038          NaN
7039          NaN
7040          NaN
7041          NaN
7042          NaN

[10 rows x 33 columns]
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
df.shape
```

```
(7043, 33)
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
df.columns
```

```
Index(['CustomerID', 'Count', 'Country', 'State', 'City', 'Zip Code',
       'Lat Long', 'Latitude', 'Longitude', 'Gender', 'Senior Citizen',
       'Partner', 'Dependents', 'Tenure Months', 'Phone Service',
       'Multiple Lines', 'Internet Service', 'Online Security',
       'Online Backup', 'Device Protection', 'Tech Support', 'Streaming TV',
       'Streaming Movies', 'Contract', 'Paperless Billing', 'Payment Method',
       'Monthly Charges', 'Total Charges', 'Churn Label', 'Churn Value',
       'Churn Score', 'CLTV', 'Churn Reason'],
      dtype='object')
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
print(df.info)
```

```
7042  3186-AJIEK      1  United States  California  Apple Valley     92308

                   Lat Long   Latitude   Longitude  Gender  ...  \
0     33.964131, -118.272783  33.964131 -118.272783    Male  ...
1      34.059281, -118.30742  34.059281 -118.307420  Female  ...
2     34.048013, -118.293953  34.048013 -118.293953  Female  ...
3     34.062125, -118.315709  34.062125 -118.315709  Female  ...
4     34.039224, -118.266293  34.039224 -118.266293    Male  ...
...                      ...        ...         ...     ...  ...
7038  34.341737, -116.539416  34.341737 -116.539416  Female  ...
7039  34.667815, -117.536183  34.667815 -117.536183    Male  ...
7040  34.559882, -115.637164  34.559882 -115.637164  Female  ...
7041     34.1678, -116.86433  34.167800 -116.864330  Female  ...
7042  34.424926, -117.184503  34.424926 -117.184503    Male  ...

           Contract Paperless Billing             Payment Method  \
0     Month-to-month               Yes               Mailed check
1     Month-to-month               Yes           Electronic check
2     Month-to-month               Yes           Electronic check
3     Month-to-month               Yes           Electronic check
4     Month-to-month               Yes  Bank transfer (automatic)
...              ...               ...                        ...
7038        Two year               Yes  Bank transfer (automatic)
7039        One year               Yes               Mailed check
7040        One year               Yes    Credit card (automatic)
7041  Month-to-month               Yes           Electronic check
7042        Two year               Yes  Bank transfer (automatic)

      Monthly Charges Total Charges Churn Label Churn Value Churn Score  CLTV  \
0               53.85        108.15         Yes           1          86  3239
1               70.70        151.65         Yes           1          67  2701
2               99.65         820.5         Yes           1          86  5372
3              104.80       3046.05         Yes           1          84  5003
4              103.70        5036.3         Yes           1          89  5340
...               ...           ...         ...         ...         ...   ...
7038            21.15        1419.4          No           0          45  5306
7039            84.80        1990.5          No           0          59  2140
7040           103.20        7362.9          No           0          71  5560
7041            29.60        346.45          No           0          59  2793
7042           105.65        6844.5          No           0          38  5097

                      Churn Reason
0     Competitor made better offer
1                            Moved
2                            Moved
3                            Moved
4     Competitor had better devices
...                            ...
7038                           NaN
7039                           NaN
7040                           NaN
7041                           NaN
7042                           NaN

[7043 rows x 33 columns]>
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
print(df.describe())
```

```
        Count      Zip Code     Latitude    Longitude  Tenure Months  \
count  7043.0   7043.000000  7043.000000  7043.000000    7043.000000
mean      1.0  93521.964646    36.282441  -119.798880      32.371149
std       0.0   1865.794555     2.455723     2.157889      24.559481
min       1.0  90001.000000    32.555828  -124.301372       0.000000
25%       1.0  92102.000000    34.030915  -121.815412       9.000000
50%       1.0  93552.000000    36.391777  -119.730885      29.000000
75%       1.0  95351.000000    38.224869  -118.043237      55.000000
max       1.0  96161.000000    41.962127  -114.192901      72.000000

       Monthly Charges  Churn Value  Churn Score         CLTV
count      7043.000000  7043.000000  7043.000000  7043.000000
mean         64.761692     0.265370    58.699418  4400.295755
std          30.090047     0.441561    21.525131  1183.057152
min          18.250000     0.000000     5.000000  2003.000000
25%          35.500000     0.000000    40.000000  3469.000000
50%          70.350000     0.000000    61.000000  4527.000000
75%          89.850000     1.000000    75.000000  5380.500000
max         118.750000     1.000000   100.000000  6500.000000
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
print(df.isnull().sum())
```

```
CustomerID              0
Count                   0
Country                 0
State                   0
City                    0
Zip Code                0
Lat Long                0
Latitude                0
Longitude               0
Gender                  0
Senior Citizen          0
Partner                 0
Dependents              0
Tenure Months           0
Phone Service           0
Multiple Lines          0
Internet Service        0
Online Security         0
Online Backup           0
Device Protection       0
Tech Support            0
Streaming TV            0
Streaming Movies        0
Contract                0
Paperless Billing       0
Payment Method          0
Monthly Charges         0
Total Charges           0
Churn Label             0
Churn Value             0
Churn Score             0
CLTV                    0
Churn Reason         5174
dtype: int64
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
print(df.drop_duplicates())
```

```
4205  9097-ZUBYC      1  United States  California  North San Juan    95960.0
4206  5148-ORICT      1  United States  California      Olivehurst    95961.0
4207  4893-GYUJU      1  United States  California    Oregon House    95962.0
4208  4578-PHJYZ      1  United States         Cal             NaN        NaN

                    Lat Long   Latitude   Longitude  Gender  ...  \
0     33.964131, -118.272783  33.964131 -118.272783    Male  ...
1      34.059281, -118.30742  34.059281 -118.307420  Female  ...
2     34.048013, -118.293953  34.048013 -118.293953  Female  ...
3     34.062125, -118.315709  34.062125 -118.315709  Female  ...
4     34.039224, -118.266293  34.039224 -118.266293    Male  ...
...                      ...        ...        ...     ...  ...
4204  39.333737, -120.858667  39.333737 -120.858667    Male  ...
4205  39.423046, -120.984472  39.423046 -120.984472    Male  ...
4206   39.082568, -121.55325  39.082568 -121.553250  Female  ...
4207  39.342587, -121.249833  39.342587 -121.249833  Female  ...
4208                     NaN        NaN         NaN     NaN  ...

            Contract Paperless Billing             Payment Method  \
0     Month-to-month               Yes               Mailed check
1     Month-to-month               Yes           Electronic check
2     Month-to-month               Yes           Electronic check
3     Month-to-month               Yes           Electronic check
4     Month-to-month               Yes  Bank transfer (automatic)
...              ...               ...                        ...
4204        One year                No               Mailed check
4205  Month-to-month                No  Bank transfer (automatic)
4206        Two year               Yes               Mailed check
4207        One year               Yes               Mailed check
4208             NaN               NaN                        NaN

      Monthly Charges Total Charges Churn Label Churn Value Churn Score  \
```

```
4206           74.35        4759.55          No         0.0        80.0
4207           19.35           1033          No         0.0        39.0
4208             NaN            NaN         NaN         NaN         NaN

        CLTV                   Churn Reason
0     3239.0   Competitor made better offer
1     2701.0                          Moved
2     5372.0                          Moved
3     5003.0                          Moved
4     5340.0  Competitor had better devices
...      ...                            ...
4204  3000.0                            NaN
4205  2168.0                            NaN
4206  4781.0                            NaN
4207  5247.0                            NaN
4208     NaN                            NaN

[4209 rows x 33 columns]
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
num_cols = df.select_dtypes(include=np.number).columns
for col in num_cols:
    df[col] = df[col].fillna(df[col].median())

for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
print(df.isnull().sum())
```

```
CustomerID           0
Count                0
Country              0
State                0
City                 0
Zip Code             0
Lat Long             0
Latitude             0
Longitude            0
Gender               0
Senior Citizen       0
Partner              0
Dependents           0
Tenure Months        0
Phone Service        0
Multiple Lines       0
Internet Service     0
Online Security      0
Online Backup        0
Device Protection    0
Tech Support         0
Streaming TV         0
Streaming Movies     0
Contract             0
Paperless Billing    0
Payment Method       0
Monthly Charges      0
Total Charges        0
Churn Label          0
Churn Value          0
Churn Score          0
CLTV                 0
Churn Reason         0
dtype: int64
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
le=LabelEncoder()
df["Gender"] = le.fit_transform(df["Gender"])
df["Partner"] = le.fit_transform(df["Partner"])
df["Dependents"] = le.fit_transform(df["Dependents"])
df["Contract"] = le.fit_transform(df["Contract"])
df["Payment Method"] = le.fit_transform(df["Payment Method"])
df["Internet Service"] = le.fit_transform(df["Internet Service"])
print(df.head(1))
```

```
   CustomerID  Count        Country       State         City  Zip Code  \
0  3668-QPYBK      1  United States  California  Los Angeles     90003
```

```
                Lat Long   Latitude   Longitude  Gender  ... Contract  \
0  33.964131, -118.272783  33.964131 -118.272783       1  ...        0

   Paperless Billing  Payment Method  Monthly Charges Total Charges  \
0                Yes               3            53.85        108.15

  Churn Label  Churn Value Churn Score  CLTV                   Churn Reason
0         Yes            1          86  3239  Competitor made better offer

[1 rows x 33 columns]
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, StandardScaler
df=pd.read_csv("/content/Telco_customer_churn.csv")
df["Total Charges"] = pd.to_numeric(df["Total Charges"], errors="coerce")
df["Total Charges"] = df["Total Charges"].fillna(df["Total Charges"].median())
df["Monthly Charges"] = df["Monthly Charges"].fillna(df["Monthly Charges"].median())
df["Tenure Months"] = df["Tenure Months"].fillna(df["Tenure Months"].median())

scaler = StandardScaler()

df[["Tenure Months", "Monthly Charges", "Total Charges"]] =
    scaler.fit_transform( df[["Tenure Months", "Monthly Charges", "Total Charges"]]
)
print(df[["Tenure Months", "Monthly Charges", "Total Charges"]].head())

plt.figure(figsize=(6,4))
sns.histplot(df["Monthly Charges"], kde=True, bins=30)
plt.title("Distribution of Monthly Charges") plt.show()


plt.figure(figsize=(6,4))
sns.histplot(df["Tenure Months"], kde=True, bins=30)
plt.title("Distribution of Tenure")
plt.show()

plt.figure(figsize=(7,5))
sns.countplot(x="Contract", hue="Churn Value", data=df)
plt.title("Contract vs Churn")
plt.show()

plt.figure(figsize=(6,5))
sns.boxplot(x="Churn Value", y="Monthly Charges", data=df)
plt.title("Monthly Charges by Churn")
plt.show()

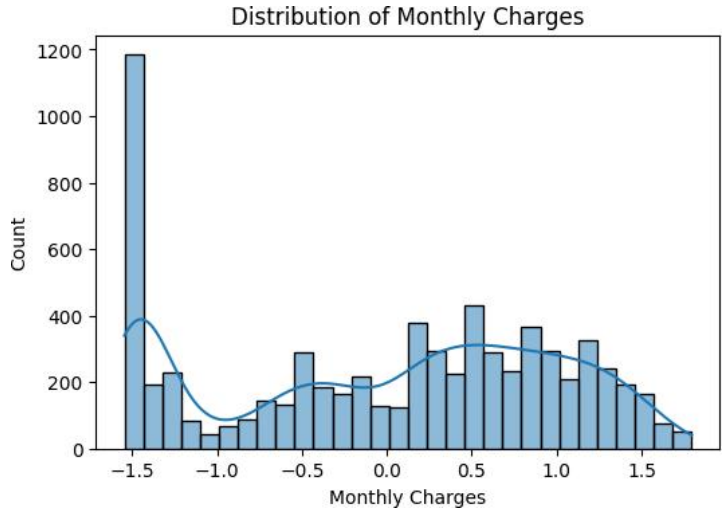

plt.figure(figsize=(12,8))

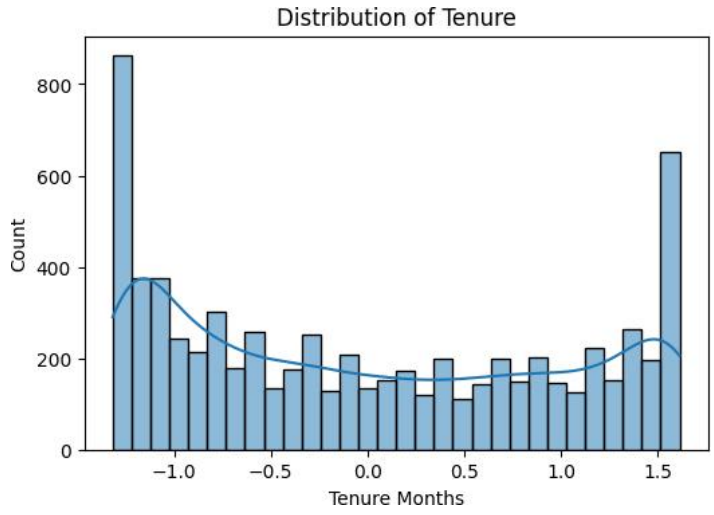
numeric_df = df.select_dtypes(include=np.number)

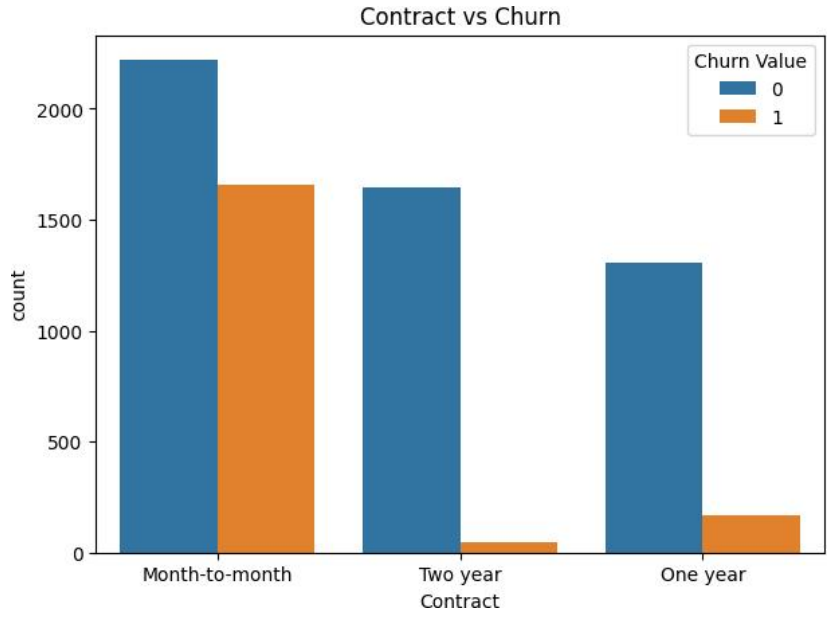
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

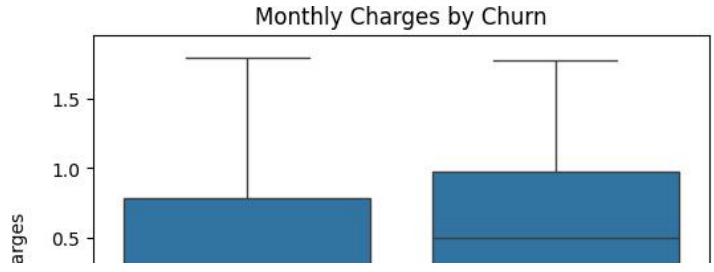
plt.title("Correlation Heatmap")
plt.show()
```

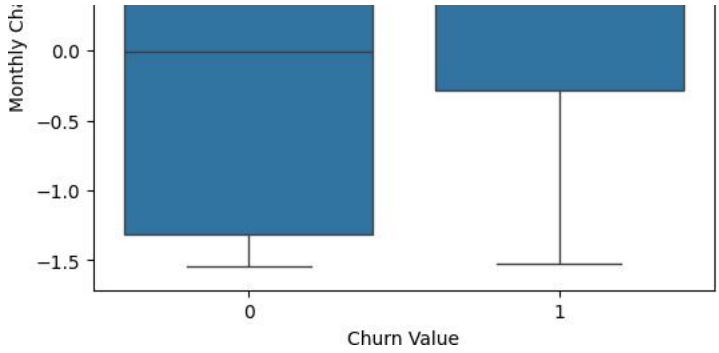
```
   Tenure Months  Monthly Charges  Total Charges
0      -1.236724        -0.362660      -0.959674
1      -1.236724         0.197365      -0.940470
2      -0.992402         1.159546      -0.645186
3      -0.177995         1.330711       0.337349
4       0.677133         1.294151       1.216004
```

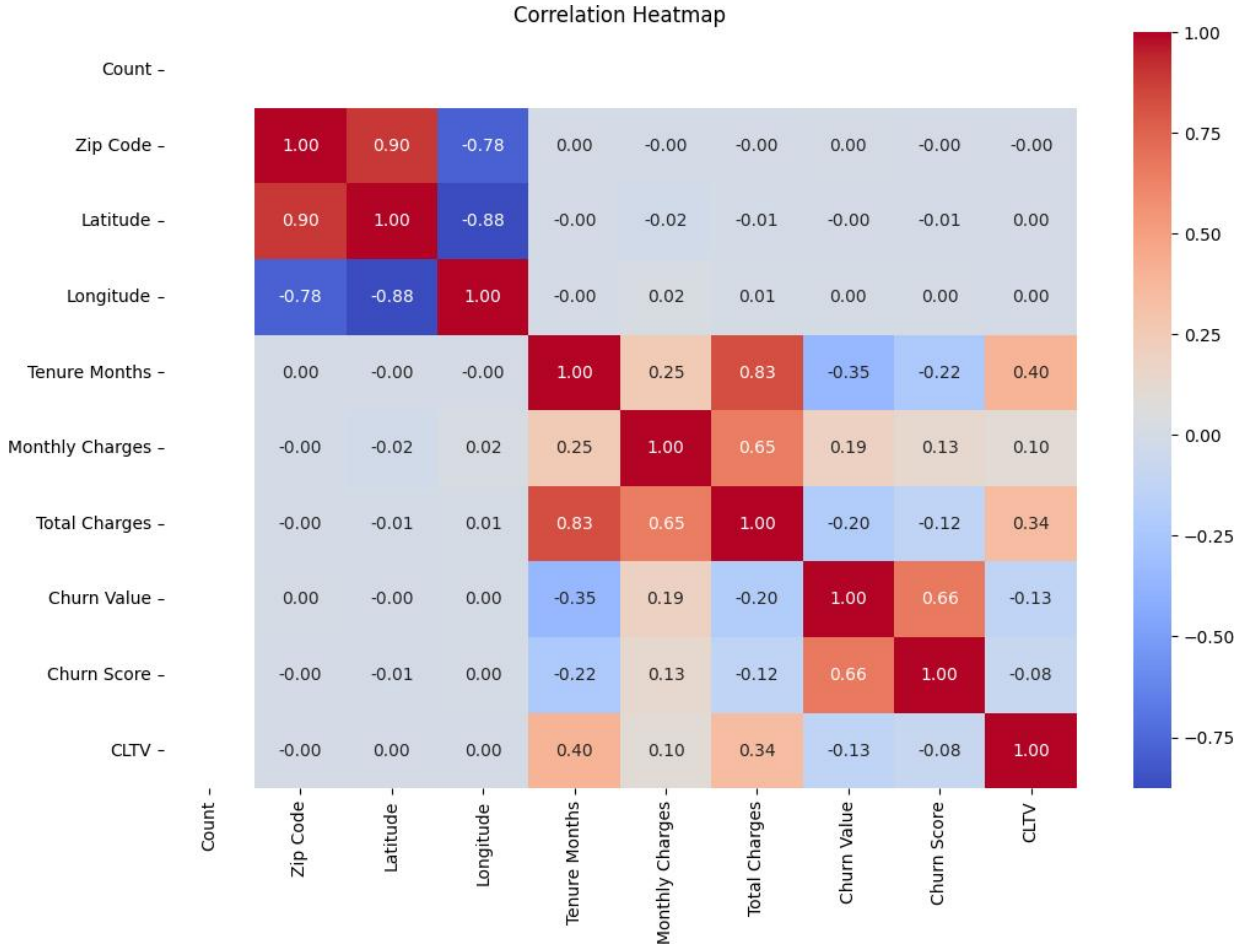









Monthly Ch
0.0
−0.5
−1.0
−1.5
0
1
Churn Value


Correlation Heatmap
Count
Zip Code
Latitude
Longitude
Tenure Months
Monthly Charges
Total Charges
Churn Value
Churn Score
CLTV
1.00 0.90 -0.78 0.00 -0.00 -0.00 0.00 -0.00 -0.00
0.90 1.00 -0.88 -0.00 -0.02 -0.01 -0.00 -0.01 0.00
-0.78 -0.88 1.00 -0.00 0.02 0.01 0.00 0.00 0.00
0.00 -0.00 -0.00 1.00 0.25 0.83 -0.35 -0.22 0.40
-0.00 -0.02 0.02 0.25 1.00 0.65 0.19 0.13 0.10
-0.00 -0.01 0.01 0.83 0.65 1.00 -0.20 -0.12 0.34
0.00 -0.00 0.00 -0.35 0.19 -0.20 1.00 0.66 -0.13
-0.00 -0.01 0.00 -0.22 0.13 -0.12 0.66 1.00 -0.08
-0.00 0.00 0.00 0.40 0.10 0.34 -0.13 -0.08 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75